

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Artificial Intelligence (AI) and Data Science) SEMESTER - 6 Winter 2025 (Supply.)

Course :Bachelor of Technology (Artificial Intelligence (AI) and Data Science) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTAIOE604A - BIG DATA ANALYTICS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12**
- 1 What are the main components of big data?
A .HDFS B. Map reduce C. YARN D. All of the above
 - 2 Data in _____ bytes size is called big data
A. Meta B. Giga C. Tera D. Peta
 - 3 Fixed-size pieces of MapReduce job is known as _____
A. Splits B. tasks C. Maps D. Records
 - 4 Which node in HDFS is responsible for managing the filesystem namespace and metadata (e.g., file names, directory structure, and block locations)?
A) DataNode B) Client Node C) ResourceManager D) NameNode
 - 5 What is the default block size for a file in modern Hadoop versions (3.x and later)?
A) 64 MB B) 128 MB C) 256 MB D) 512 MB
 - 6 HDFS ensures a high degree of fault tolerance by:
A) Using expensive, highly reliable hardware.
B) Storing all data on a single, centralized server.
C) Replicating data blocks across different DataNodes
D) Performing real-time data analysis.
 - 7 In the MapReduce job submission process in YARN, which component is responsible for launching the ApplicationMaster?
A) JobClient B) ResourceManager
C) NodeManager D) TaskTracker
 - 8 The fundamental unit of data processed by MapReduce is:
A) A single record. B) A dataset. C) A key/value pair. D) A file.
 - 9 _____ is a component on top of Spark Core.
A) Spark Streaming B) Spark SQL C) RDDs D) All of the mentioned
 - 10 _____ leverages Spark Core fast scheduling capability to perform streaming analytics.
A) MLlib B) Spark Streaming C) GraphX D) RDDs

- 11 _____ is a distributed machine learning framework on top of Spark.
 A) MLlib B) Spark Streaming C) GraphX D) RDDs
- 12 GraphX provides an API for expressing graph computation that can model the _____ abstraction.
 A) GaAdt B) Spark Core C) Pregel D) None of the mentioned
- Q2. Solve the following.
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| A) Discuss about Big-Data in detail | 6 |
| B) Explain in detail, Apache Hadoop and Hadoop Echo System | 6 |
- Q3. Solve the following.
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| A) Explain the concept of HDFS in detail | 6 |
| B) Illustrate the Hadoop core components with neat diagram | 6 |
- Q4. Solve Any Two of the following.
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| A) Explain the Concept of Map reduce with proper example | 6 |
| B) Discuss about Map Reduce types and Hadoop cluster | 6 |
| C) Discuss about RDBMS and Big SQL | 6 |
- Q5. Solve Any Two of the following.
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| A) What is PIG in Big Data? Difference between Pig and Database | 6 |
| B) Illustrate main features and Architecture of Hive with neat diagram | 6 |
| C) Write a short notes on Hbase, Hive QL | 6 |
- Q6. Solve Any Two of the following.
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| A) Explain in detail Apache Spark with architecture | 6 |
| B) Discuss about Apache Kafka concept and architecture | 6 |
| C) Discuss about Kerberos authentication with architecture | 6 |

***** End *****